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Math 140 Calculus I with Precalculus, Part 1 Spring 2025
Section 01

Class Time: M/W/F, 9:00am-9:50am
Classroom: Arter Hall 105

Instructor: Hang Zhao
Email: hzhao@allegheny.edu
Office hours:

- M/F 11:00am-12:00pm, 2:25pm-3:25pm Location: Alden Hall 105
- Tue 12:00pm-1:00pm. Location: Google Meet
- W 11:00am-12:00pm. Location: Alden Hall 105

By appointment at: https://calendar.app.google/eq6qaUi8CLVmpXCDA

Availability: My office hours are "open door" -- no appointment is needed. In addition to office hours, I am available at other times. I am usually available immediately after class (including my last class of the day – see my schedule). I can meet with you at other times, but you'll need to contact me ahead of time to set an appointment. I shall provide you my schedule in my first email. It is also available in the Google Drive Folder for your course.

Communication: Students are expected to check their Allegheny email regularly (at least twice per day). The best way to reach Prof. Hollerman is via email.

Attendance: If you must miss a class, please send me an email *before* class explaining your need to be absent. You are then responsible for catching up on the work that you missed and for getting any important information that was shared in class. Your first resource for this is other students in the class. College faculty do not provide lectures for individual students who miss class.
It will be assumed that you have either print or electronic access to the textbook during class.

Course Description: Credits: 4 An introduction to the differential and integral calculus of algebraic functions, the natural logarithmic function, and the natural exponential function, including limits, derivatives and their applications, integrals, and the Fundamental Theorem of Calculus. Review of topics from precalculus is integrated throughout the calculus material.
This course is not intended for students who have MATH 151 with a grade of C or better.

Distribution Requirement: QR

Calculator: You will need a non-graphing calculator that includes function keys for exponents, e^x, and logarithms, $ln\ ln\ x$. For timed assessments, you may not use a calculator on a multi-purpose device (phone, tablet, etc.). Those who plan to continue into Math 141 might need a calculator with trigonometric functions for that class.
These calculators will be fine for students in my Math 140 and Math 141 classes. They are available from Walmart and from Amazon among many other places.

Texas-Instruments-TI-30XA-Student-Scientific-Calculator (can be found for under $15)
Casio-FX-260-Solar-II-Scientific-Calculator (also can be found for under $15)

Learning Goals: Students who are successful in this course will be able to

1. demonstrate knowledge of mathematical content involving limits, differentiation and integration
2. use calculus to solve applied problems and to present the solution clearly
3. write complete and correct proofs of calculus results.

Text: Textbook chapters and handouts will be provided by the instructor. It will be assumed that you have electronic access to the textbook during class.

Grading: Your success with the learning goals will be assessed using *mastery grading*.
There are several components to this, one of which is a sequence of *timed assessments*. These will allow you to demonstrate your mastery of the learning goals. More information on the grading system will follow.

Final exam: The last timed assessment will be offered at your assigned final exam time. (Assigned by the college.)
Exam group A: Monday Dec 9 9 am – 12 pm

Except in emergencies, your final exam time cannot be changed the last week of class.
If you need to request a different time, please contact me well before the end of the semester. I'll need time to get your request approved by me and by the Registrar.

Honor Code:
It is imperative that students understand the Honor Code. You have a responsibility to practice academic honesty at all times, and to take appropriate action if you know of a violation of the Honor Code.
For more information on the Honor Code and on General College Policies, see *The Compass Student Handbook*.
In order for your timed assessment to be graded, it must include the pledge: "This work is mine unless otherwise cited" and your signature.
You may work with others in *preparing* for assessments and in solving workbook problems.

You may neither give nor receive aid on any assignment to be graded for this class.

You may not submit materials from this course to depositories maintained by any group or organization or individual who is not currently enrolled in this section of Math 140. Any such submission will be in violation of the Honor Code,

Allegheny College Statement of Community: Allegheny students and employees are committed to creating an inclusive, respectful and safe residential learning community that will actively confront and challenge racism, sexism, heterosexism, religious bigotry, and other forms of harassment and discrimination. We encourage individual growth by promoting a free exchange of ideas in a setting that values diversity, trust and equality. So that the right of all to participate in a shared learning experience is upheld, Allegheny affirms its commitment to the principles of freedom of speech and inquiry, while at the same time fostering responsibility and accountability in the exercise of these freedoms. This statement does not replace existing personnel policies and codes of conduct.

Disability Services: Students with disabilities who believe they may need accommodations in this class are encouraged to contact Student Disability Services as soon as possible to ensure that accommodations are approved and implemented in a timely fashion. The phone number is (814) 332-2898. Disability Services is located in the Learning Commons on the first floor of Pelletier Library.

The grading of the course is described on the next few pages.

Grading the course Spring 2025

This section of Math 140 will be graded by your demonstration of mastery of various course objectives. Every individual evaluation is done using pass or no pass. So, for example, on a particular attempt, you have either found the equation of a line correctly or you have not.

There are four broad kinds of objectives.
Specific Content Objectives 1 are subdivided into four categories.
Objective 2 requires you to choose a method appropriate to a particular function.
Objective 3 involves mathematics and writing assignments.

You will have opportunities to earn “pass” on Objectives of type 1 and 2 on in-class timed assessments.
Timed Assessments:
The in-class timed assessments will be offered on the following Fridays: Jan 31, Feb 14, Feb 28, Mar 14, Mar 28, and Apr 18. For this class, assessments end at 9:55 am.
You will not be expected to complete all problems on each assessment. Instead, you'll select from the problems, the ones that correspond to objectives that you still need to pass and that you think you can do correctly. If your answer is completely correct, you'll receive a Pass for the corresponding objective. If your answer is not completely correct, you'll not receive a Pass and you will have opportunities to reattempt that objective on a later assessment. (Exception: the last chance is on the final assessment.)
Final exam: The last assessment will be offered at your final exam time. (Assigned by the college.)
Exam group A: Thursday May 1 9 am – 12 pm

You may also take advantage of an **Office** attempt to pass the objective.

Office Attempts: You may attempt to earn a pass for an objective during office hours or during an appointment you make with me. The number of objectives you may attempt each week depends on how many you have attempted so far in the semester. Everyone begins at level 1.

- Level 1 You may make one attempt to pass an objective, per week.
- Level 2 You may make two attempts to pass objectives per week. You enter level 2 when you have made 4 attempts at objectives in the office. (whether you passed or not.)
- Level 3 You may make three attempts to pass objectives per week. You enter level 3 when you have attempted 12 objectives in the office. (whether you passed or not.)
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This system is deliberately designed to reward students who begin working on objectives early in the semester.

Note re: Office Attempts: I encourage you to visit my office when you have questions about the material. I am happy to help you learn this subject. Note, however, that If you ask me questions about a course objective, you may **not** make an office attempt later that day. You'll need to make your attempt on a later day.

The last Office attempts will be offered on Wednesday Apr 30 (study day).

A list of the objectives follows.

Specific Content Objectives 1: (There are four categories of content objectives.)

	Precalculus Topics (PC) (Requires one Pass to Master)	Mastered
PC.1	I can determine the equation of a linear function. (The equation of a line.)	
PC.2	I can find the average rate of change of a function on an interval.	
PC.3	I can use the laws of exponents to convert to/from negative integer exponents.	
PC.4	I can use the laws of exponents and convert between rational exponents and radicals	
PC.5	I can simplify an expression involving exponents.	
PC.6	I can solve a quadratic equation using factoring.	
PC.7	I can solve a quadratic equation using the quadratic formula.	
PC.8	I can solve equations involving polynomials (by factoring).	

	Graphs and Calculus (GC) (Requires one Pass to Master)	Mastered
GC.1	Given the graph of a function, I can identify the graph of its derivative.	
GC.2	I can find limits of a function using graphical methods.	
GC.3	I can find the equation of the tangent line to a function at a given point.	
GC.4	I can find the critical numbers and critical points of a polynomial function.	
GC.5	I can determine where a function is increasing and decreasing.	
GC.6	I can determine the intervals of concavity of a function and find all of its points of inflection.	
GC.7	I can apply the First Derivative Test to classify the critical points of a function.	
GC.8	I can apply the Second Derivative Test to classify the critical points of a function.	

	Calculus Skills (CS) (Requires one Pass to Master)	Mastered
CS.1	I can compute the derivative of a function using the limit definition.	
CS.2	I can differentiate combinations of power functions.	
CS.3	I can apply the Product Rule to differentiate functions.	
CS.4	I can apply the Quotient Rule to differentiate functions.	
CS.5	I can differentiate functions involving the natural exponential function.	
CS.6	I can differentiate functions involving the natural logarithmic function.	
CS.7	I can apply the Chain Rule to differentiate functions.	
CS.8	I can find the most general antiderivative of a function.	
CS.9	I can find compute definite integrals using the Fundamental Theorem of Calculus.	
CS.10	I can find antiderivatives using substitution.	
CS.11	I can find compute definite integrals using substitution.	

	Interpretation	Pass 1	Pass 2	Pass 3	Mastered

I.1	I can interpret the meaning of a function and its derivative in the context of an application using correct units of measurement.				

Course Objective 2: Determining differentiation methods for given functions and differentiating.

Given 4 functions, I can decide which differentiation rule or rules to use to differentiate them, and then differentiate the functions.

	D	C	B	A
Objective 2	1 correct ☐	2 correct ☐	3 correct ☐	4 correct ☐

Writing Assignments (Objective 3):

By mastering the objectives listed above, you will demonstrate a foundational knowledge of these concepts and skills. The writing assignments will require you to communicate this knowledge.

Course Objective 3: Problem writing assignments (PW) and Analysis (A)

	Problem Writing Assignments (WA)	Completed
PW.1	Problem Write Up 1	
PW.2	Problem Write Up 2	

	Analysis (A)	Completed
WA.1	Analysis 1	
WA.2	Analysis 2	

For grading, PW and WA Objectives will be combined into a single requirement. To earn a D in the course, you must complete one of these successfully. To earn a C in the course, you must complete two of these. For a B, you must complete three of them. In order to earn an A in the course, you must complete four successfully.

The writing assignments will be graded on a Complete/Not Complete basis. In order to earn credit for an assignment, it must be both correct and properly formatted. A correctly formatted assignment submission should look like a short essay, effectively integrating the mathematical work with explanation and reasoning. It should meet all of the following specifications:

1. The solution or proof is neatly hand-written or typed, using complete sentences. In particular, all mathematical notation and expressions should be part of a sentence.
2. The solution or proof is written at a level that is appropriate for a Math 140 audience, namely, members of our class who are familiar with the content of the course but who may not have worked on the particular problem whose solution you are presenting.
3. The solution or proof is correct and the steps in the solution are also correct.
4. The solution or proof is complete, meaning that a member of the standard Math 140 audience (see above) can trace your reasoning from beginning to end and be persuaded that your answer is correct without having to reconstruct or guess at significant portions of your thinking.

5. Your writing is almost free, if not entirely free, of spelling errors.
6. Your writing is almost free, if not entirely free, of basic grammatical errors such as incomplete sentences, subject-verb disagreement, and misuse of punctuation.
7. Mathematical notation and terminology is used properly.

The writing assignments will each have an attached rubric listing the specifications above.
If your submission meets all of these specifications, then I will mark it as Complete. Otherwise, I will use the rubric to indicate which specifications you have not yet met.
The assignments will have 3 submission deadlines. You may make one submission by each deadline. If you miss the first deadline, you only get 2 attempts to complete the assignment and so on.
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There is a one page summary of the grade requirements on the next page.

GRADE REQUIREMENTS

	D	C	B	A
Precalculus	☐☐ ☐☐	☐☐	☐	☐
Graphing and Calculus	☐☐ ☐☐	☐☐	☐	☐
Calculus Skills	☐☐ ☐☐	☐☐ ☐	☐☐	☐☐
Interpretation (each box is 1 pass)	1 correct ☐	2 correct ☐	3 correct ☐	4 correct ☐

Additional Outcomes.

	D	C	B	A
Objective 2 (number correct on a single attempt)	1 correct ☐	2 correct ☐	3 correct ☐	4 correct ☐
Objective 3	1 correct ☐	2 correct ☐	3 correct ☐	4 correct ☐

So, for example, to earn a C in the course, you need to pass 6 Precalculus and 6 Graphing and Calculus Objective. And also, you need to master (2 passes) 7 Calculus Skills Objectives. Finally, you'll need to pass 2 questions for Objective 2 and correctly present solutions to 2 problems in Objective 3.

To earn a B in the course, you need to pass 7 Precalculus and 7 Graphing and Calculus Objective. And also, you need to master 9 Calculus Skills Objectives. Finally, you'll need to pass 3 questions for Outcome 2 and correctly present solutions to 3 problems in Objective 3.